Haoyu Zhai (翟浩宇)

Email: zhaihaoyu007@gmail.com
Phone: +86 130 0252 3469
Website: hiyotaka.github.io
Google Scholar: fB7o1RoAAAAJ
GitHub: github.com/Hiyotaka
Address: Beijing, China

RESEARCH INTERESTS

Trustworthy Machine Learning; LLM Agent Security and Reliability; Out-of-Distribution / Novelty Detection; Clustering and Anomaly Detection.

EDUCATION

Shanghai Normal University 2022.09 - 2025.06 | Shanghai, China

Master of Engineering in Computer Science and Technology
Supervisor: Prof. Yan Ma
Master thesis: “Research on a Split–Merge Clustering Algorithm Based on Minimum Spanning Forests”
Major courses: Digital Image Processing (90), Digital Signal Processing (90), Computer Network and Simulation (94)

Nanjing University of Aeronautics and Astronautics Jincheng College 2017.09 - 2021.06 | Jiangsu, China

Bachelor of Engineering in Transportation
Major courses: Linear Algebra (93), Advanced Mathematics (98), Probability Theory and Mathematical Statistics (100)

PUBLICATIONS

[1] **Haoyu Zhai**, Jie Yang, Haotao Guo, Bin Wang, Yan Ma. “Density-increment and cut-edge optimized clustering via minimum spanning forest.” **Neurocomputing**, 2026. **[First author | JCR Q1]**. DOI: https://doi.org/10.1016/j.neucom.2026.132957. [code]

[2] Zexuan Fei, **Haoyu Zhai**, Jie Yang, Bin Wang, Yan Ma. “Discovering generalized clusters with adaptive mixture density-based clustering.” **Knowledge-Based Systems**, 2025. **[Co-first author | JCR Q1]**. DOI: https://doi.org/10.1016/j.knosys.2025.113250. [code]

[3] **Haoyu Zhai**, Zexuan Fei, Yan Ma. “A Novel Outlier Detection Algorithm Based on Symmetry and Distance Ratio.” **International Conference on Pattern Recognition**, 2024. **[First author]**. DOI: https://doi.org/10.1007/978-3-031-78192-6_22. [code]

RESEARCH EXPERIENCE

Multi-Tool LLM Agent System for Malware and Security Analysis 2025.07 - Present | 360 Group, Beijing

Role: Main developer / AI Algorithm Engineer

- Built a multi-tool LLM agent platform for malware/security triage, orchestrating Python, Node.js, shell execution, file operations, binary/PE inspection, and Office/PDF analysis in Docker sandboxes.
- Implemented batch multi-agent dispatch and a FastAPI WebUI for file upload, task submission, live status tracking, checklist progress, and Markdown report rendering.
- Added structured action-observation traces and repeatable experiment flows to support reliability analysis, cross-tool consistency checks, and runtime failure detection for agent systems.

NSFC Project: Graph-Based Clustering and Image Recognition 2022.09 - 2025.06 | Shanghai Normal University

Role: Algorithm designer / core contributor

- Designed graph-based clustering and outlier-detection algorithms, including symmetry-distance outlier scoring and a minimum-spanning-forest split/merge clustering framework.
- Evaluated SMMSF on 12 two-dimensional and 15 real-world datasets against 9 algorithms; used AC, PR, RE, and F1 metrics and achieved best results on 13 real-world datasets.
- Evaluated the outlier detector on 4 synthetic and 16 UCI datasets against 5 mainstream detectors; used denoising-before-clustering experiments to verify clustering gains.
- Contributed to related patent/application materials from the project under the supervision of senior collaborators.

AWARDS AND HONORS

Outstanding Graduate Award, Shanghai Normal University, 2025 (top 5%). **Shanghai Normal University Scholarship**, 2022, 2023, 2024. **NUAA Jincheng College Scholarship**, 2017, 2018, 2019, 2020.

SKILLS

- **Research:** Trustworthy ML, LLM agent reliability/security, clustering, anomaly detection, OOD/novelty detection, computer vision.
- **Engineering:** Python, C/C++, Matlab, PyTorch, Scikit-learn, OpenCV, Docker, FastAPI, Linux, Git, Milvus, LaTeX/Typst.
- **Languages:** Mandarin Chinese (native), English (working proficiency).